

Quanteon Solutions AI/ML Engineer - What to Learn

MUST Learn (Class A keywords with no direct project experience)

- **Classical ML stack (NumPy, Pandas, Scikit-learn)** – your genuine AI experience is at the LLM/application layer (RAG, agents, tool calling), not classical ML model training. NumPy/Pandas are data manipulation libraries; Scikit-learn covers traditional ML algorithms (regression, classification, clustering). Do a focused course or Kaggle notebook covering data preprocessing through model evaluation to have a concrete example to discuss.
- **Deep learning frameworks (TensorFlow, PyTorch, Keras)** – frameworks for building and training neural networks from scratch, a different skill from calling a hosted LLM API (which is what Echo/Vibe do). PyTorch is generally considered the more research-friendly, Pythonic choice; Keras (now part of TensorFlow) is the higher-level, faster-to-prototype option. Build one small image classifier or text classifier end-to-end to understand the training loop (forward pass, loss, backward pass, optimizer step).
- **Computer vision & NLP model training** – distinct from prompt-engineering an existing LLM. Computer vision covers CNNs, image classification/detection; NLP model training (as opposed to using a pretrained LLM) covers tokenization, embeddings, and fine-tuning transformer models. This is likely the largest gap versus your current AI experience.
- **LLM fine-tuning** – you’ve used pretrained LLMs via API (Gemini, OpenAI) extensively, but haven’t fine-tuned a model’s weights yourself. Look at LoRA/QLoRA (parameter-efficient fine-tuning) as the practical, resource-light entry point rather than full fine-tuning.

GOOD to Learn (strengthens the application, not blocking)

- **MLOps tooling (MLflow, Airflow, DVC, Kubeflow)** – experiment tracking, pipeline orchestration, and data versioning for ML workflows. You have CI/CD pipeline experience (Clink AI, GitHub Actions) – the concepts (automation, reproducibility) transfer, but these are ML-specific tools you haven’t used.
- **Big data technologies (Spark, Hadoop)** – distributed data processing frameworks for datasets too large for a single machine. No direct analogue in your current experience; worth a conceptual overview (what problem they solve) even without hands-on practice.
- **Cloud ML services (AWS SageMaker, GCP Vertex AI)** – managed platforms for training and deploying ML models. You have AWS experience (Secrets Manager in Echo) but not the ML-specific services; skim SageMaker’s basic workflow (notebook -> training job -> endpoint) since it’s explicitly named as a preference.

Fast-Ramp Note

Your genuine AI strength is at the application/agent layer – RAG pipelines, LLM tool calling, prompt engineering, vector search, and voice AI (Echo, Vibe) – which covers this JD’s ”generative AI, voice AI, prompt engineering, RAG, vector databases, agent AI” line directly. The classical ML/DL stack (NumPy/Pandas/Scikit-learn/TensorFlow/PyTorch/Keras, computer vision, model fine-tuning) is a genuinely separate skill set from LLM application development, and is where most of your prep time should go if you pursue this role – it’s less a ramp than a second specialization.